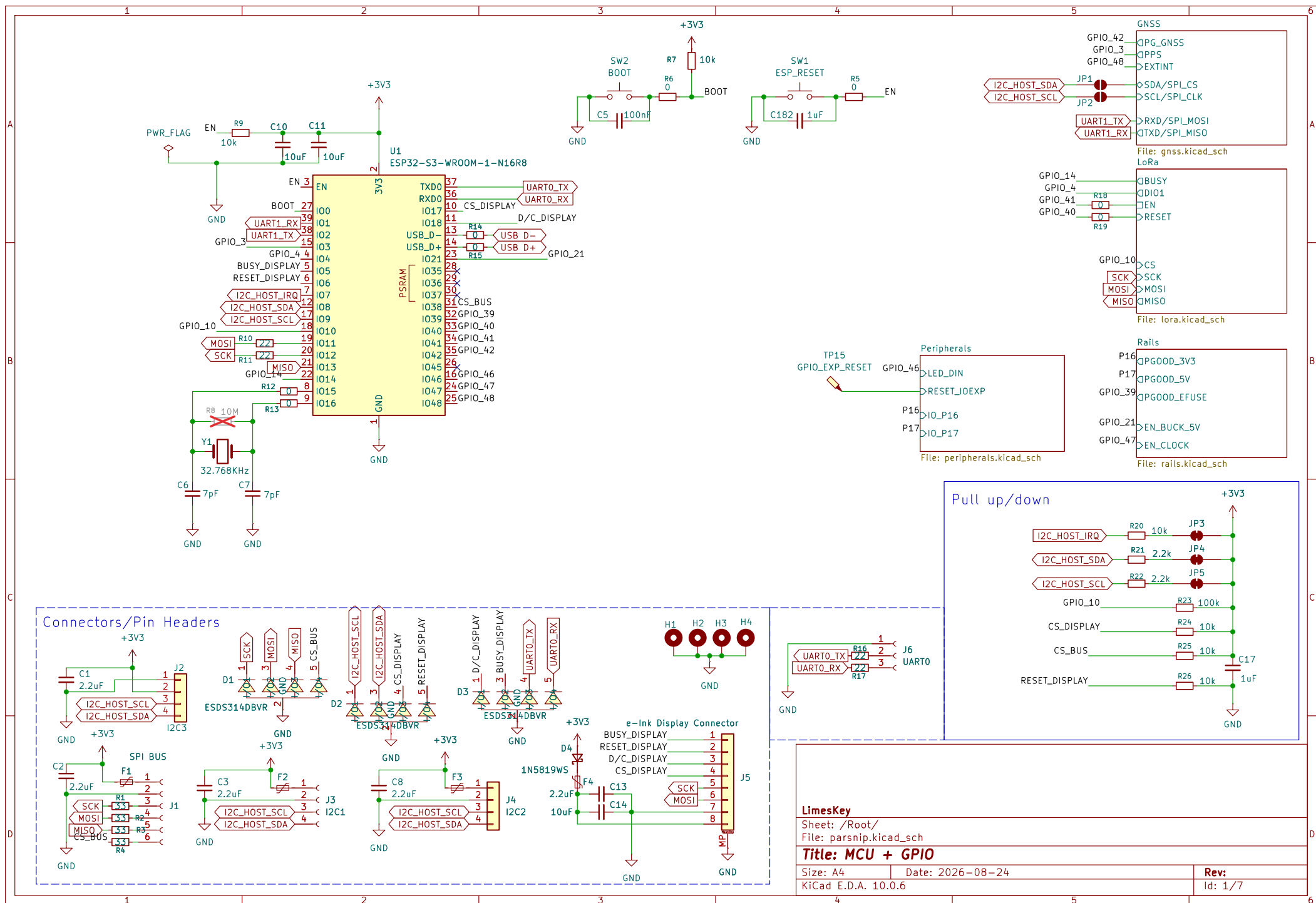




[illegible]

3V3 Buck

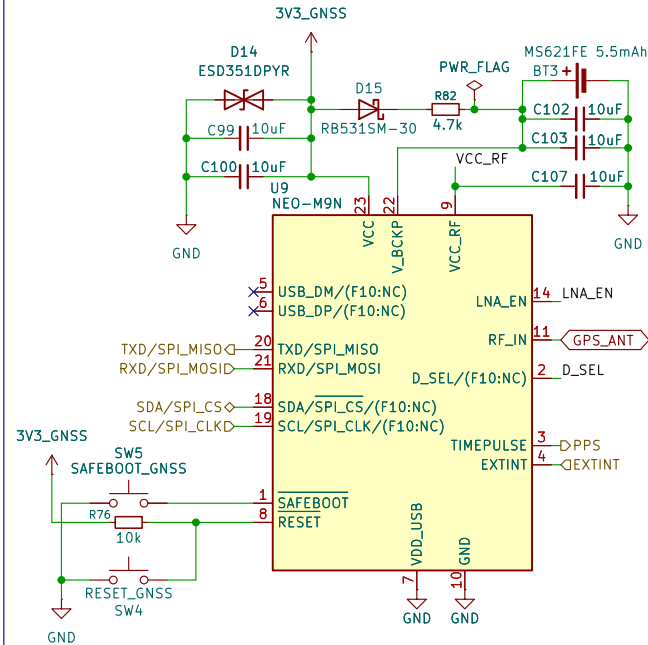
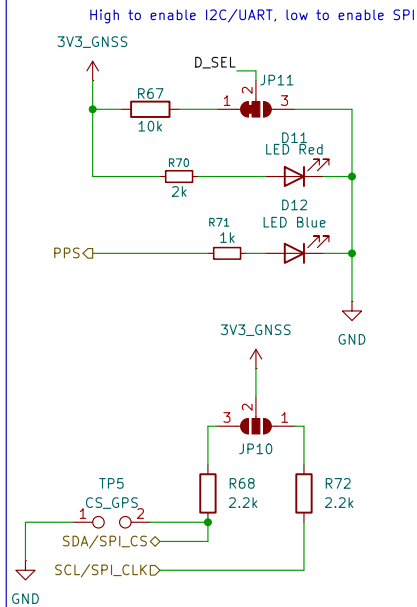
The schematic shows the 3V3 Buck converter circuit. The IC is the LM61460AASQRJRRQ1. The input voltage is VSYS, connected to VIN1 (pin 8). The input filter consists of capacitors C122, C123, and C124. The feedback network includes R109 (6.04k) and C125 (2.2uF) connected to the EN/SYNC pin (pin 7). The switching network includes the SW pin (pin 10) connected to the inductor L4 (2.2uH) and capacitors C131 (100nF), C133 (220pF), C135 (2.2uF), C141 (2 x 10uF), and C143 (22uF). The output filter includes capacitors C190, C191, and C192. The output voltage is +3V3. The PWR_FLAG signal is shown as an output. The circuit is powered by VSYS and GND.

Optional: Buck's VIN 180 degrees out of phase with eachother for less EMI

4MHz / 2 = 2MHz each buck

Rev:
Id: 4/7

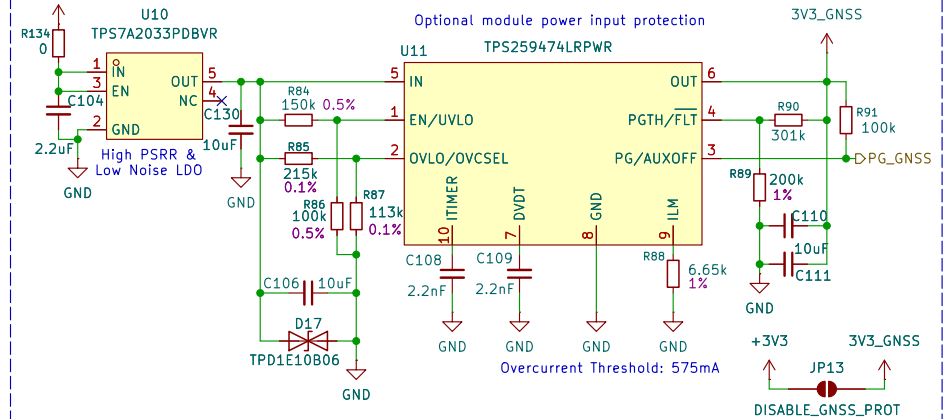
LEDs and Jumpers



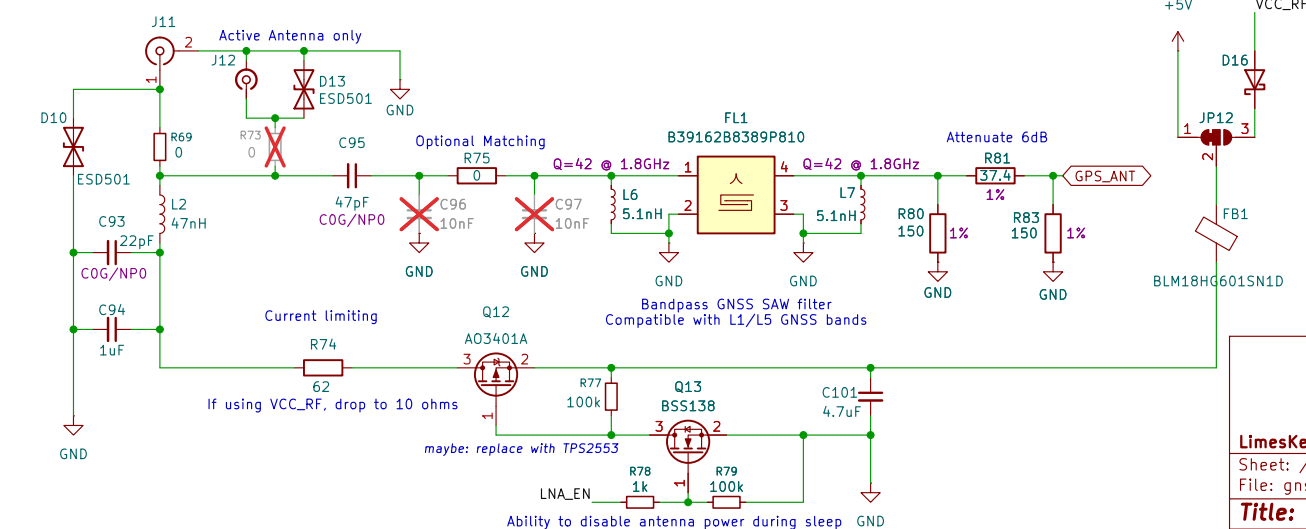
Pin-Pin compatible with NEO-M9N/M8N/M7N/M6N & NEO-F10N
The NEO-F10N only supports UART

Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf

Optional LDO & eFuse



SMA + IPEX Connector



LimesKey

Sheet: /Root/GNSS/
File: gnss.kicad_sch

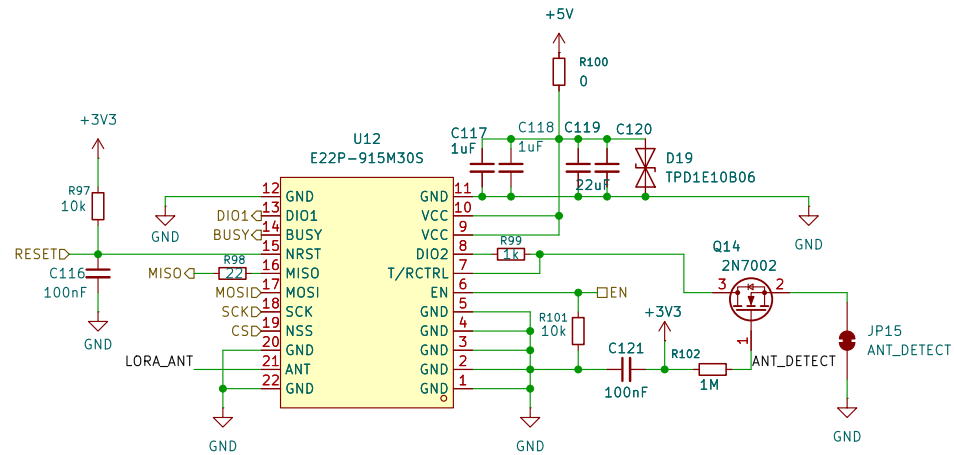
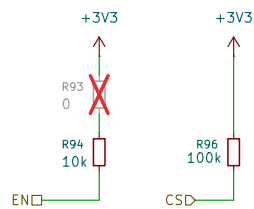
Title: GNSS Module / Active RF Antenna

Size: A4
KiCad E.D.A. 10.0.6

Date: 2026-09-14

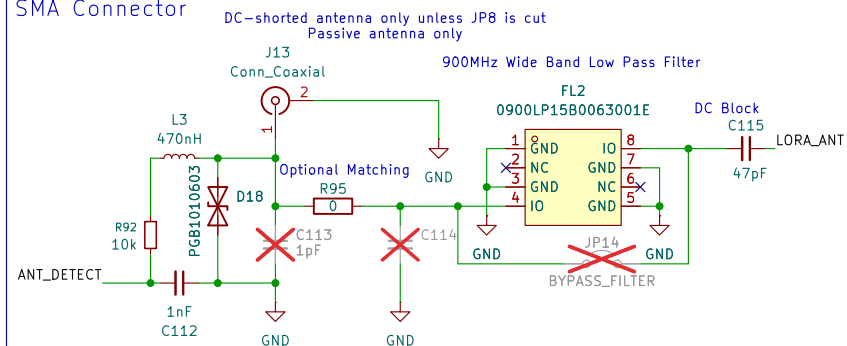
Rev: 1.1

Id: 5/7



Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4ffaf

SMA Connector



LimesKey

Sheet: /Root/LoRa/
 File: lora.kicad_sch

Title: LoRa Module / LoRa Antenna

Size: A4 Date: 2026-08-24

KiCad E.D.A. 10.0.6

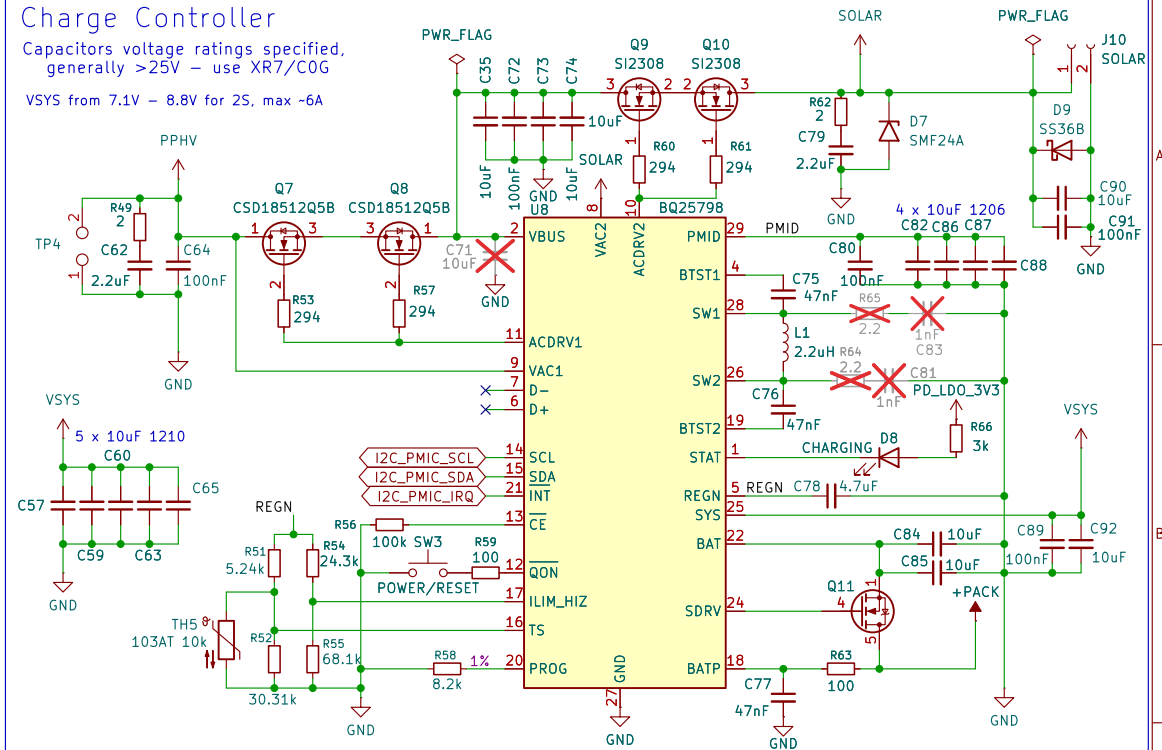
Rev:
 Id: 6/7

For use with WS2813/WS2816

Charge Controller

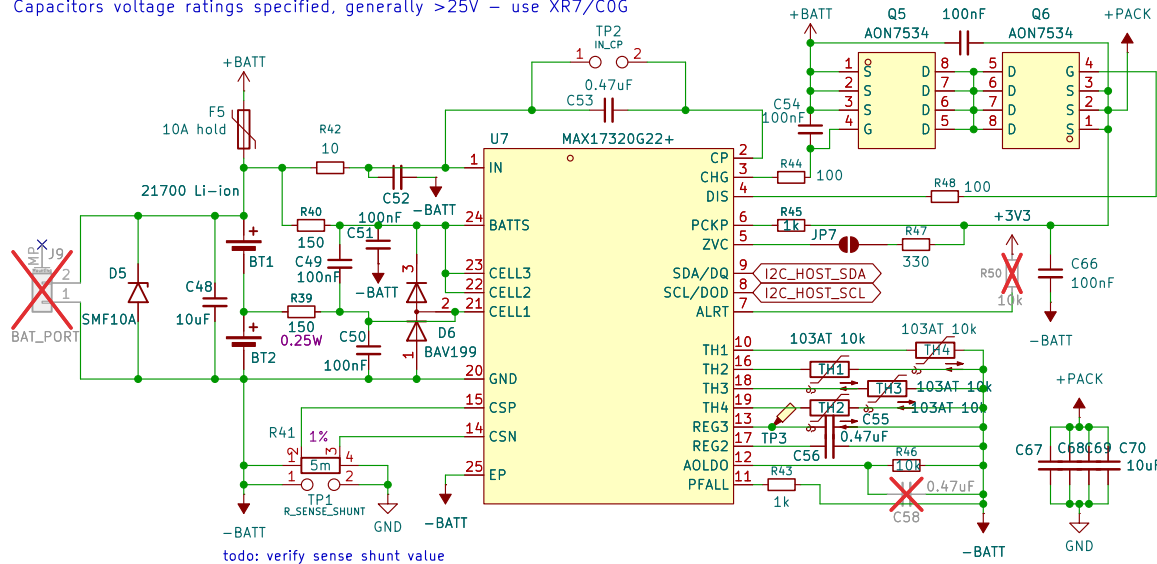
Capacitors voltage ratings specified, generally >25V – use XR7/COG

VSYS from 7.1V – 8.8V for 2S, max -6A



Battery Management System (BMS)

Capacitors voltage ratings specified, generally >25V – use XR7/COG



LimesKey

Sheet: /Charger/
File: battery.kicad_sch

Title: Battery Charger / Battery Management System (BMS)

Size: A4 Date: 2026-08-24

KiCad E.D.A. 10.0.6

Rev:
Id: 3/7